

1(a)(i)	metaphase ;	1
1(a)(ii)	telophase ; I <i>ref. to</i> early / late	1
1(b)(i)	osmosis ; <i>any two from:</i> water (molecules) move, down a water potential gradient / from high(er) to low(er) water potential / from less negative to more negative water potential ; A ψ for water potential I <i>ref. to</i> concentration gradient across the cell (surface) membrane ; passively / passive movement / no energy required / no ATP required ;	3
1(b)(ii)	single (partially permeable) membrane / tonoplast ; A membrane-bound / bound by a membrane R envelope / membranes / double-membrane bound R if tonoplast is also described as a wall contains <u>cell sap</u> or detail ; e.g. (solution) containing dissolved (organic and inorganic) substances / AW water and other substances	2
1(c)	xylem drawn as cross or star <u>and</u> phloem drawn in areas between xylem ; R if vascular bundles around the edge drawn centrally within whole section of root ;	2

Question	Answer	Marks
1(d)	any four from: 1 (because) presence of, sucrose / assimilates, causes water potential (inside phloem sieve tube) to, be lower / decrease / AW ; I water potential <u>gradient</u> decreases 2 water enters, phloem (vessels) / sieve tubes, down a water potential gradient / by osmosis ; I down a concentration gradient I higher water potential gradient to lower water potential gradient 3 increase in volume ; <i>in context of within sieve tubes, at source / in roots</i> 4 (so) increase in / high(er), <u>hydrostatic</u> pressure ; <i>must have hydrostatic once</i> A builds up I creates / forms 5 unloading / removal, of, sucrose / assimilates, at, sink / growing areas, lowers <u>hydrostatic</u> pressure ; <i>context is water follows osmotically</i> 6 (mass flow / bulk flow / movement) down pressure gradient / from high(er) to low(er) pressure ;	4

Question	Answer	Marks
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